

Securities Education Certificate



Exclusive SEC Cartoon by Nicolas Geiseler Toran (Mechanical Engineering, 2022) [1/7]

Lecture 1

An introduction to the finance industry (industry background)

Lecturers

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Introduction

A Simplistic Example of the Stock Market

A security is a tradable asset or certificate that has monetary value. What do I mean by this? Well imagine your friend has started a company and is looking for money. As a good friend and self-proclaimed opportunistic investor, you decide to invest £100 of your hard earned money into his company. In exchange, you receive a certificate saying that you now own 1% of his company. Now, you can do 1 of 2 things:

1. **Keep your share of the company.** You invested in the company because you thought it was a good investment at £100 for a 1% share of the company. As your friend expands and grows his company, other investors may look at it and want to invest in it. The higher demand for shares will mean that investors will have to pay more for the same 1% stake. As such, the “market value”, i.e. how much people are willing to pay for your share of the company will have increased. Similarly, if people don't believe in what your friend's company is doing, the lack of demand will mean the ‘market value’ of your 1% share will drop as fewer people are willing to buy your share or are only willing to buy in at a lower price.
2. **Sell off your share of the company (i.e. trade it).** If you don't believe in what your friend's company is doing you can try to sell off your share of the company. However, depending on what other people think of your friend's company you may or may not have a hard time finding someone else willing to buy your share of the company.

What is clear from this example, is that the certificate you are currently holding, guaranteeing you 1% of the company has a monetary value. This is what is meant by a **security**.

Asset Classes

In general, there are 4 main types of investments that are tradable in markets. Each classification is known as an asset class made up of groups of investments that share common characteristics:

Equities

These are shares of companies that are traded on public exchanges. These exchanges are market places that facilitate the buying and selling of shares between 2 parties. In the example above, we briefly discussed what the market value of a security means. Well, it is the job of the stock exchange to keep up to date with what the market value of a share currently is for companies that are listed on the exchange, which in effect is how much other investors are willing to pay a share of a company and how much you would have to pay to purchase that share. If you ever decided to sell the shares of your friend's company, the stock exchange would help you to find a buyer and the price the sale would take place at, based on investor demand for your friend's company's shares, acting like a middle man in the transaction. Examples of these include the New York Stock Exchange (NYSE), London Stock Exchange (LSE), NASDAQ (predominantly comprised of tech stocks). Owning a share of a company represents owning a fraction of that company so if, say Company X has 10,000,000 shares outstanding and you buy 1 share of that company, then you own 0.00005% of the company. Depending on the

number of people that trade them, these stocks can also trade Over-The-Counter. This means that the trade is settled between the two parties, without the supervision of an exchange.

Fixed Income (Bonds)

Bonds are debt securities that are issued by both corporations and governments. These are essentially IOUs (“I Owe yoU”), where companies and governments borrow money (the total amount borrowed is called the principal) from lenders for a fixed period of time (e.g. up to 3, 5 or even 10 years) at a fixed coupon rate (the amount of yearly interest the borrower will pay the lender). At the end of the contract, i.e. at maturity, the principal amount is paid back to the lender. Therefore, if a lender holds a bond until maturity, they will receive both their principal back and interest that they have earned from the borrower for lending out their money.

Investors holding bonds do not have to hold them to maturity, and can trade them either on public exchanges or Over-The-Counter (OTC). Do not worry about the technical details discussed and how bonds are valued - this will be covered in Lecture 6. The important point here is to note that bonds are one of the main securities that can be traded in markets.

As an example, let's say you lend your friend, Bobby, £20,000 (principal) over 5 years to buy a car and who agrees to pay 1% (coupon rate) of the principal back to you every year. So every year, Bobby pays you £200, until the end of year 5 (maturity), when he returns you the original £20,000. One thing to note is that you could have sold the bond before maturity.

Foreign Exchange (FX)

The Foreign Exchange market is the largest market in the world worth US\$1.93 quadrillion. Each day, US\$5.3 trillion is traded. More than 85% of the market⁽¹⁾ features transactions between the 7 major currency pairs which are (EUR/USD, USD/JPY, GBP/USD, AUD/USD, NZD/USD, USD/CAD, USD/CHF). What are currency pairs? Currency pairs allow us to compare the value of one currency to another - the first currency listed is called the base currency and the second currency listed is the quote currency. The base currency is the currency you want to buy, while the quote currency is the currency you want to sell. Essentially, say you are holding JPY but believe that the USD will increase in value. Why? Several factors may come into play such as politics where the country might be undergoing elections, or an important piece of economic data has just been released detailing that the economy of the USA is growing better than expected. You then go to your broker who will give you a 2-way quote on the USD/JPY currency pair (note that USD is the base currency, i.e. the one you want to buy), sell him/her your JPY and receive USD in exchange. If you're confused by currency pairs... Don't worry! Our Lead Lecturer Ahmad will be guiding you through the entirety of FX in Lecture 2, but I've given you an introductory scope of the topic and what you'll be finding out soon when he takes over next time.

The FX market is considered an OTC market in the sense that there is no centralised exchange that oversees the trading of the currency pairs in circulation, and the exchange runs 24h a day for 5 days a week.

Commodities

These include physical resources such as Oil & Gas, precious metals such as Gold, Silver, metals such as Lithium, Copper that are essential for electric vehicles. Even wheat, grain, coffee and beans are considered commodities. As highlighted by the supply chain shocks due to the war in Russia and Ukraine, the prices of these resources are determined by the supply-demand dynamics which are affected by factors such as

weather, geopolitical tensions, trading routes etc (Will be covered more in Lecture 2). One way for investors to invest in this asset class is to buy financial instruments called futures, which are contracts where the investor has agreed to buy/sell a specific amount of the commodity at a predetermined price at a specific time in the future. These are traded on futures exchanges. Again, do not worry this will be covered more in-depth in Lecture 7 by Joseph.

Over recent years, the emergence of cryptocurrencies have prompted debates about where it sits as an asset class i.e. whether it should be considered a commodity or an asset class on its own given its role as both a store of value and use as currency. We'll leave you to decide where you would class them!

Industry Players

Although it might seem confusing at first, the finance industry can be split broadly into 4 separate categories – of which we will go into more detail below:

Sell-Side	Buy-Side
Regulators	Central Bank

Sell-side

These are predominantly the investment banks that you are probably familiar with. Think Goldman Sachs, JP Morgan, Morgan Stanley etc. These banks are classified as financial services companies that act typically as advisors to clients from the buy-side and take a fee from the services they provide, but do not invest any of their own funds i.e. from their balance sheet into the market. Think of them as the gate-keepers to the financial markets, helping companies raise money or facilitate acquisitions/mergers between companies.

Buy-side

As its name suggests, firms on the buy-side invest their own money or clients' money into different asset classes (i.e. they buy assets usually by enlisting the help of the sell-side). What they invest in, and how long they choose to invest defines the type of company they are classified as. This includes companies such as Private Equity Funds, Asset Managers, Venture Capitalists and Hedge funds.

Regulators

This third group is involved in ensuring that the market participants are operating fairly to prevent market manipulation. Market manipulation refers to when a market participant (a bank, hedge fund, investors, company chairmen etc.) artificially tries to increase or decrease the stock price of a company under false pretences. A great example of this is the “Pump-and-Dump” scheme that you might have heard of that featured in *The Wolf of Wall Street*. Essentially, Jordan Belfort led a stock broking firm that held a large amount of shares in questionable penny stocks - stocks that are not traded around as much (we call this low liquidity). The firm would call up potential investors and feed them false stories about these companies to make them buy shares in the penny stocks the firm owned, causing the share prices of these companies to rise (the pump), before exiting their positions (the dump) and leaving investors facing losses.

In the USA, the Securities and Exchange Commission (SEC) is in charge of enforcing securities law to prevent cases like this from occurring. In the UK, this is the Financial Conduct Authority (FCA).

The Central Bank

Typically, each country has its own central bank – the USA has the federal reserve or the 'Fed', while the UK has the Bank of England. The banks do not function like how a normal bank does – they exist as a separate entity from the government and act with the primary goal of controlling inflation (i.e. the total amount of money in circulation in the economy) by adjusting interest rates on the Federal Funds Rate. This is the rate that commercial banks lend to each other (mostly overnight). These banks have to borrow from each other as they must all have enough reserves (the cash they are holding) to make up a percentage of customer deposits they hold. Why does this matter? This is the base rate that all other rates are determined from, so government bonds, company bonds, lending rates including mortgages are all adjusted based on this rate. If the Federal Funds Rate is increased, this will mean that banks will also raise their own interest rates when they lend out their money.

It is very important to control inflation as if more money is in circulation in the economy, the prices of goods and services will increase resulting in a rising cost of living. The interest rate is the single most important number in the world of finance as all other lending rates depend on it. If the Fed sets a higher interest rate, it means borrowing costs will go up for corporations as well as consumers. As a result, corporations are less likely to borrow money and subsequent business and economic activity will decrease resulting in a fall in the total amount of money in circulation. We have seen especially this year that the Fed has been very aggressively trying to raise interest rates in order to combat inflation, so now you know why they have to do this! We will cover this topic much more in depth in Macroeconomics in lecture 2.

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Scroll on over to the next page to find out more about the sell-side.

The Sell-Side

Now we will go into detail on the Sell-side. As mentioned above, the sell-side consists of investment banks.

Structure of a typical investment bank

An investment bank is typically split into 3 main areas. The Front Office which features the Investment Banking and Markets divisions, the Middle Office, which includes Risk Management and Compliance, and Back Office, which includes other support functions such as Human Resources and Technology.

The Front Office is further split into two sides - the public facing side, featuring the markets division and equity research teams and the private side, which features the Investment Banking division. Very often, Sales & Trading is classified as an Investment Banking activity but falls under the markets side of the bank. Typically, Investment Banking refers to the advisory (with respect to raising capital or a transaction between two companies) and underwriting services (left) that banks provide to their clients. We will go into this in more detail below.

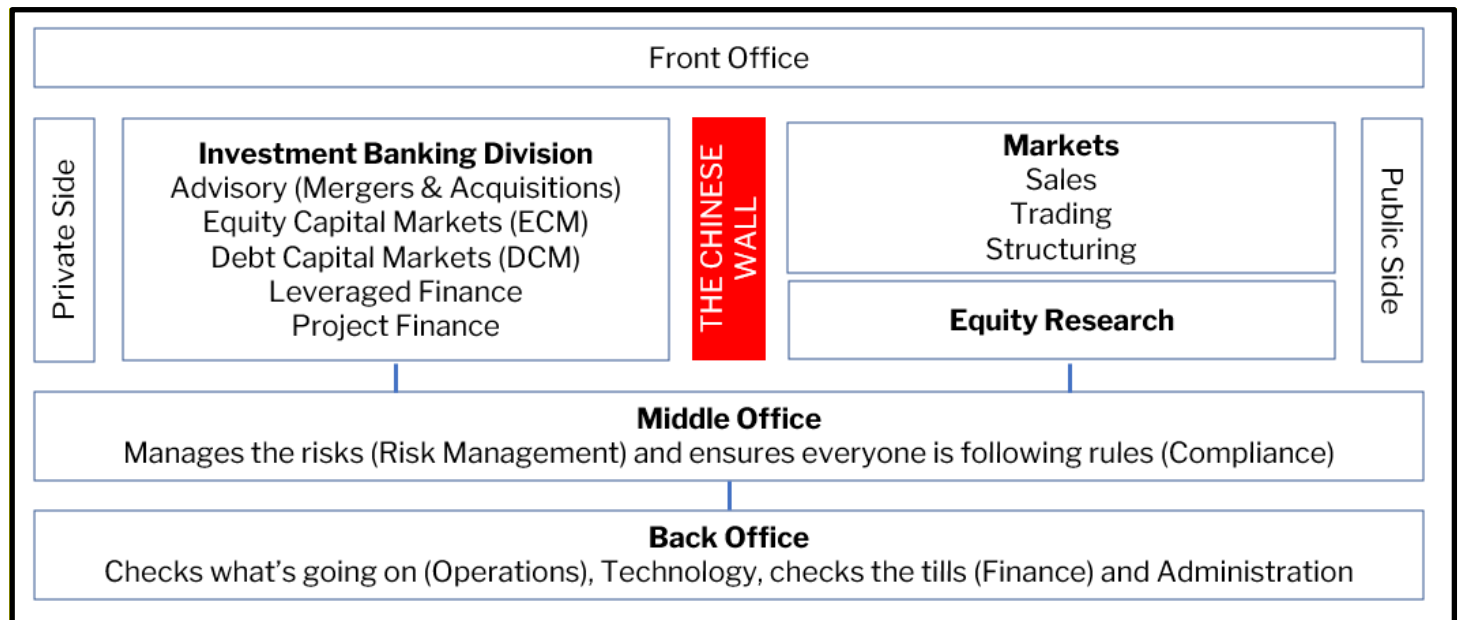


Figure 1: Structure of a BB Investment Bank

Investment Banking

This typically includes the bank's advisory side of the business. This is known as the private side of the bank because you are dealing with deals that are not yet known to the market. On this side of the business you will find the following divisions:

M&A deal teams

These teams do what is most commonly perceived as Investment Banking. The teams typically cover different sectors such as Healthcare, Technology, Media and Telecommunications (TMT), Infrastructure, Consumer and Retail and advise on both buy-side and sell-side deals i.e. advising the buyer in a deal or advising a seller on a sale.

For example, imagine you are a managing director at Jefferies in the Healthcare sector. From your research you have identified that Pfizer could be in the market to acquire a smaller pharmaceutical company specialising in immuno-oncology treatments. You then analyse potential acquisition targets by analysing their business and valuing them. You pitch this to Pfizer and then execute the deal. This is a simple example of a buy-side deal. On the opposite side, there will be another investment bank acting as advisors to the smaller oncology drugs company who will be facilitating its sale.

Equity Capital Markets (ECM)

ECM focuses on servicing companies that are interested in raising capital through equity – to put it simply, this means it services companies who want to raise money by issuing new shares to the stock markets. What is Capital? Well, capital is nothing more than money that companies raise in order for them to either fund their strategy by maybe building more factories, investing in research and development or even acquiring another company. So in effect, when you buy these new shares, you are directly providing funding for these companies to carry out their specific expansion plans!

This is typically done in 3 different ways:

Initial Public Offerings (IPO)

This is the most well-known process of raising capital. When a private company (i.e. owned by only a select list of investors including founders) has grown to a certain size, they will want to raise capital to continue funding their expansion plans. They therefore decide to sell a proportion of their company to the public markets where the company's stock will be traded. They will of course retain a majority shareholding so that they can retain control of company decisions.

Typically in an IPO, a group of banks (usually 4-5) will work together to carry out the transaction. This group is known as the syndicate, and their primary job is to find investors to invest in the IPO company. Very often, the syndicate will underwrite the deal. What does this mean? In every IPO process, the banks will work together to value the IPO company. For example, let's say Uber wants to IPO and is valued at £1bn. **Underwriting** this means that the syndicate of banks will guarantee that Uber will receive £1bn of capital regardless of whether the banks find enough investors to invest £1bn in total. So the banks take on the risk of the IPO not raising enough money and will have to pay Uber the remaining amount of money should they not be able to find enough investors for the IPO.

Of the syndicate, 1-2 of these banks will act as Global Coordinators who take a principal role in sorting out the logistics of the deal including the legal processes that must be carried out for the company to be listed, and take on a larger amount of the risk from underwriting as highlighted above. The other banks are known as bookrunners play a more secondary role and only focus on finding investors for the transaction. The process of finding investors is called **bookbuilding**.

Rights Issue (RI)

This is for companies that are already public who want to raise capital – either because they need to fund their capital expenditure plans or because they are in financial trouble and need to pay down their debts. The company typically issues new shares at a discounted price that is offered to existing shareholders to avoid dilution. (Since creating more shares will mean that all existing shareholders will own a smaller proportion of the company).

Rights Issues typically run over a few weeks as investors are typically given a week to decide if they want to subscribe to new shares or sell-off their right to buy shares in the market and face their percentage of ownership decreasing (diluted).

Accelerated Book Building (ABB)

ABBs are similar to Rights Issues – they are an alternative method for companies to raise capital. As mentioned in the IPO section, a bookbuild is when banks find investors to invest in the company in question - combining this with the word “accelerated” and the name of this transaction immediately makes sense. There are two main types: Primary ABBs and Secondary ABBs. In a Primary ABB, the company simply issues new shares and it is the bank's job to find investors that are willing to invest in these shares. In a Secondary ABB, the company in question wants to sell off shares it has in another company or a subsidiary – it is a secondary transaction because no new shares are being created.

ABBs are very fast processes that do not require much paperwork, generally taking a few hours. For this reason, there are rules put in place where the maximum number of new shares issued can only be a maximum of 20% of the current number of shares outstanding. In contrast, in a Rights Issue, typically a maximum of 50% the current number of shares outstanding can be issued to the market. So for instance if Uber has 100 million shares in total before the rights issue, it can only issue a maximum of 50 million new shares.

Debt Capital Markets (DCM)

Similar to ECM, DCM is the part of the investment bank that looks to help companies raise capital but through the issuance of debt. DCM can be split into Investment Grade bonds and High Yield bonds (aka junk bonds).

The difference between IG bonds and High Yield bonds lies in their rating. Each bond is rated by a rating agency (E.g. Moody's, S&P) based on the ability of the borrower to return the amount borrowed with interest. These ratings range from AAA to C or even D. The lower a bond is rated, the more likely the borrower will default on its repayment obligations and hence the higher the returns the investor will demand resulting in a higher coupon rate. Anything under BBB (S&P) or Baa (Moody's) is considered a junk bond.

Like ECM, analysts in this area have to be very up-to-date with the market in order for them to price the coupon rates of the bonds issued appropriately.

Leveraged Finance

Leveraged Finance is the part of the IB that services Private Equity firms, also known as financial sponsors. As we will see below, Private equity firms seek to acquire companies through a mixture of debt and equity, meaning that they both borrow money and invest their own money when acquiring another company. Their goal is to improve the operations of the target company over a period of time and use the earnings generated by the company to pay down the debt they have taken on to purchase the company.

In this division, analysts create Leveraged Buy-Out (LBO) models in order to predict the cash flows of the company that is to be acquired in order to determine how much they are willing to lend to the financial sponsor and the interest rate they will charge. Typically, there is a consortium of banks called a syndicate that provide the loans in order to spread the risk of lending to the sponsor.

Depending on the Investment Bank, this division could cover high-yield bonds that companies can use to raise capital. High yield bonds simply mean that the interest the company has to pay on them is typically higher due

their perceived riskiness. Factors such as the amount of debt a company has taken on, how much profit they are making and how well people think the company will do are all taken into account.

Project Finance

Project Finance (PF) is very similar to leveraged finance – the only difference is that instead of dealing with the acquisition of companies, PF deals with real-assets that are typically in the renewable/infrastructure finance, for example, offshore wind farms. Clients can be either financial sponsors or developers, and projects can be built from scratch (greenfield) or re-developed (brownfield).

Types of Investment Banks

Bulge Brackets (BB)

These banks are the biggest players in the industry, rank the highest in the IB league tables servicing the largest clients and working on the largest deals.

e.g. Goldman Sachs, JP Morgan, Morgan Stanley, UBS, Credit Suisse.

Middle Market Players

These banks are structured in the same way as BBs but typically service smaller clients. They are typically Universal Banks – banks that provide the whole spectrum of services ranging from retail banking (where you store your money), commercial banking (providing loans to clients), investment banking, asset management and private banking.

e.g. HSBC, Standard Chartered

Elite Boutiques

These are pure play investment banks that focus only on advising clients. These firms do not underwrite their transactions, and do not possess a Markets Division (i.e. Sales and Trading). This is meant to make their advice less biased as they will only receive fees for advising and not from underwriting and selling products to the market.

e.g. Lazard, Rothschild, PWP, Evercore.

Markets

The markets division at an Investment Bank encompasses the following roles:

1. Sales
2. Trading
3. Structuring
4. Equity Research

Contrary to the private side where the Investment Banking division sits, the markets division sits on the public side as the information they deal with is all public which everyone has access to. This division between the

public and private sides of the business is called '**The Chinese Wall**' meaning that there can be no communication between both sides of the bank. This is very tightly regulated by compliance in the bank as there is a big conflict of interest between the public and private sides of a bank. As you can imagine, if any private-side information from the investment banking side is leaked to the sales and trading desk, the sales and trading desk will be able to profit based on an unfair advantage compared to other investors in the market.

Sales & Trading

Sales and trading share what we call a symbiotic relationship - each one depends on the other in order to best fulfil the client's needs.

People working in Sales must have comprehensive and detailed knowledge of the products the bank offers, for example swaps, options (which we will cover in Lecture 4) etc.. So in effect, the salesperson is the person on the phone to, say, Brevan Howard, to discuss an order Brevan Howard would like to be placed. In order to effectively communicate the bid-ask spreads to Brevan Howard, they will have to be in constant contact with the trader to get this information and relay it back to the client.

There are 2 main types of trading:

1. Agency Trading
2. Market-making

Fun fact: After the Volcker Rule was passed in 2010 by then Fed chairman Paul Volcker, proprietary trading was banned in all banks as a result of the 08-09 Global Financial Crisis to prevent banks from using their own accounts to make speculative investments that might lead to similar crises. Originally, in 1933, the government introduced the Glass-Steagall act which was aimed at preventing commercial banks from investing clients' deposits into the market. This was repealed in 1999, but all it took was 10 years before it had to be reintroduced.

Agency Trading

In agency trading, you are servicing clients on the buy-side (typically hedge funds) by executing orders and trading on behalf of them. Why is this needed? Because typically these clients put in orders that are big enough to cause drastic changes in a given company's share price. As a trader, it is your responsibility to execute the order in a way that the stock price remains steady so that your client can fill their order at the lowest possible price.

For example, imagine you are an agency trader at BNP Paribas. Your sales person comes to you one day saying that Brevan Howard (a hedge fund) would like to place a big order of 500 shares of Tesla (its stock ticker – how it is recognised on the stock exchange – is TSLA) stock. Because the order amount is much bigger than a normal retail investor, it would be difficult to purchase the shares at one go without moving the share price upwards. For example, if at the time you received the order, TSLA was trading at \$700. What I mean is it is impossible for you to buy all 500 shares at \$700 as, due to the size of the order, every time you buy say 100 shares, the price of TSLA will be pushed upwards, meaning your client would have to pay a higher price. It is therefore your job as an agency trader to ensure that the order is executed in the most efficient way possible so Brevan Howard pays the best price (lowest) for their order.

Market-Making

If you have ever purchased or sold shares on the market, have you ever wondered how your order is always seamlessly and immediately filled? Imagine this, say right now you decide to buy 10 shares of TSLA. If you go to your brokerage account on Trading 212, you will realise that all you need to do is click “buy” and you will then immediately be the owner of 10 TSLA shares. This is the same vice versa. How is this possible?

If you think about it, that implies that there must be a buyer on the other end who has bought the shares you are selling and vice versa if you are selling.

This is not the case in reality.

The first thing you need to know about market-making is the bid-ask spread.

Whenever a trade is to be placed at the sales desk of the bank, the salesperson will always quote a bid/ask.

The Bid-Ask is always seen from the **bank’s perspective**. Bid in this case is the amount that the bank will **bid** to buy your shares of TSLA for when you wish to sell it, and Ask would be the amount the bank would **ask** you to pay for them to sell shares of XX company to you. The difference between both quotes is called the bid-offer spread. A spread is typically the difference between two things - in this case, it is the difference between the bid and ask prices.

$$\text{Bid-Offer spread} = \text{Bid} - \text{Ask}$$

The SPOT price, which is the actual price of the security, will lie in between the bid and ask prices. Why? If you think about it, no middle-man in a deal is going to buy at exactly at the bid price because they want to be compensated for the service they have provided by acting as market maker to allow you to sell your shares immediately. Therefore, naturally and normally, the SPOT price will lie somewhere in between the bid and ask prices so that the bank always buys at a lower price and sells at a higher price than the SPOT.

	Bid	Spot	Ask
<i>JPM</i>	92.49	92.53	92.55
<i>GS</i>	92.54	92.60	92.72
<i>BNPP</i>	91.70	92.00	92.40

It is the job of the salesperson to communicate this with the client effectively, and the job of the trader to analyse the current market conditions to provide a competitive bid-offer spread. There are tons of banks on the street that are competing for the same business, so as a trader you really want to deliver a price that is good for the bank in terms of profit, but also for the bidding client to make them feel like they have got the best deal and will not go to a competitor.

As you might have guessed, the trader profits via the bid-offer spread. In an ideal world the trader will remain delta neutral i.e. they will profit regardless of the direction the security is trading in, but only through the spread.

Let's go through an example so this is clearer:

Say you are a trader at a hedge fund and you want to sell 1000 shares of AAPL. You call a salesperson at the bank, Bobby. Bobby communicates with a trader, Sophie who quotes a two way price. You have already enquired about other banks and their spreads and have decided that this is the best deal for you (i.e. the tightest spread), so you sell your shares to the bank. Now, Sophie does not want to take any directionality on her position, but finds herself with 1000 shares of AAPL. He will then need to work together with the salesperson to both hedge (i.e. manage the risk of holding the 1000 shares - this will be covered in Lecture 4) and unwind (i.e. sell off) his position to avoid making a loss with his current position.

A Challenging Case Study – EUR/GBP Currency Pair

Hopes are high on the GBP since the markets believe a Brexit deal will soon be approved in parliament. Jasmine, a broker at Morgan Stanley, wants to be **short** on this pair unless James is willing to pay a higher price. This means Jasmine would much rather sell EUR to get GBP as opposed to buying EUR with GBP. James likes to think of himself as a successful day trader, and firmly believes that GBP will rise, meaning that he is **long** on GBP.

To avoid confusion, I want to clarify that both Jasmine and James share the same view that GBP will rise against the EUR which effectively means that they are both **short** on the EUR/GBP pair as they expect the value of EUR to drop against GBP.

On Feb 1, 2019, Jasmine gives James the following quote:

Bid (James sells EUR for GBP)	Market	Ask (James buys EUR with GBP)
£0.892	£0.900	£0.901

We can see that because Jasmine is long on GBP, she has set her bid-ask spread to be a lot more attractive to her receiving EUR - there is only a 0.001 difference between the market price and the ask price, making it much more attractive for people to sell their GBP to her. This is in contrast to the Bid price which is 0.008 away from the market price which makes it much less attractive for people to sell their EUR for GBP. As mentioned above, James is long on GBP so he sells his EUR where €1 and gets £0.892.

Two months later, the U.K. leaves the E.U. with a good Brexit deal advantageous for the economy, including a single market agreement. The market is now reflecting this outcome, so Jasmine is now **neutral** on the pair.

On Apr 2, 2019, James now wants to sell his GBP for EUR to make a profit, since the price of EUR has dropped relative to GBP. He goes to Jasmine and receives the following bid-ask spread.

Bid (James sells EUR for GBP)	Market	Ask (James buys EUR with GBP)
£0.889	£0.892	£0.896

This means if he were to sell his £0.892, he would get back $£0.892/£0.896 = €0.996$. This is €0.004 less than what he started with (€1) even though he took the correct position in believing that GBP would rise. In fact, this difference is Jasmine's profit as a market-maker!

As seen by this example, although as a trader you do not profit from the direction of the security that you are trading, you aim to use your knowledge of the market to create a bid-ask spread that maximises your profit while remaining competitive to the market.

Structuring

Structuring is the process of creating bespoke products for clients' complex needs and views on markets. What does bespoke mean here? Well, in general this involves the combination of 2 or more 'vanilla' (vanilla refers to the most basic version of a financial instrument as opposed to exotic, which represents a more complicated instrument that combines a few vanilla products) products that produce a pay-off profile (i.e. how much you can profit and how much you can lose in different scenarios - do not worry, we will be covering this in Lecture 4) that suits what a company needs. In general there are 2 types of structuring products:

1. Interest rate structuring
2. Equities derivatives structuring

The main clients of interest rate structuring are pension funds while the main clients of the equities derivatives side are hedge funds. Typically these companies come to you looking for ways to mitigate the risk of future market conditions (for example if the Fed raises interest rates) or simply to mitigate the risks of their unique investment strategies. They will usually have an idea about what this product should do, but don't have access to it through the Sales & Trading desks or the product might not exist at all. Examples of structuring products include combining products such as bonds with call options (we will cover options in Lecture 4) or multiple call/put options together.

For example, let's say you are an analyst covering credit structuring and a pension fund comes to you. They know that in the future they will have to pay out a certain amount of money to pensioners, however they do not know what the market conditions will be like (such as inflation) up to the time they make that payout. Therefore, it is your job as a structuring analyst to create a product through the combination of other products such as bonds and options and accurately price them that fits the specific situation. In general the role of a structuring analyst is a mix of getting exposure to the clients, the market and quantitative skills (for pricing).

Equity Research

Equity Research is the part of Markets that focuses on analysing companies in different sectors, and sells their research to investors such as hedge funds and investment banks that want access to their research to help with their investment ideas or to examine potential deals they want to explore. Examples of these sectors include Renewables, Healthcare, Oil & Gas, Technology, Media & Telecommunications (TMT) etc. Typically, these teams will all operate separately and cover their own sectors, and follow a specific few companies in that sector.

The role of a research analyst is to follow a particular company (or group of companies) very closely - this includes attending all of the company's announcements, speaking to their management, and attending their earnings reports every quarter. The lead analyst of the team (Managing Director) will form an opinion of the company's news and update their view on the company. Equity research analysts will split their day following these companies, speaking to investors who buy their research and pay to talk to these analysts, and writing research reports on the industry and companies. They will also be required to attend investor roadshows for IPOs which are marketing events investment banks hold with potential investors to educate them about the IPO company in order to bookbuild as we covered before.

It is very important to note that Equity Research sits on the public side. This is because their coverage of their companies should be unbiased. What do I mean? Imagine Joe is an equity research analyst at Goldman Sachs specialising in the transport sector and covering Deliveroo. One day one of his colleagues in the ECM team in the Investment Banking division of Goldman Sachs comes to him saying that he is currently working on the Deliveroo IPO. Now since this is a company Joe is covering he will be tasked to attend the investor roadshows to educate investors on the company to drum up interest in the company. There is a potential conflict of interest here as while Joe is meant to be unbiased in his opinion of the company, having very good remarks could sway investors to invest in the IPO, which would benefit the ECM team working on the deal. This is unethical and there are very strict procedures in the bank to avoid this information from crossing the private side to the public side.

The Buy-Side

Asset Management (AM)

AM services institutional investors such as pension funds, mutual funds, insurance companies by managing their funds by investing across different asset classes. AM strategies are typically Long-only and no short positions are taken.

It is common for Investment Banks such as Goldman Sachs and JP Morgan to have asset management divisions, but these typically exist as separate entities from their sell-side businesses due to potential conflicts of interest. There are also numerous well-known Asset Managers such as BlackRock and Fidelity.

Hedge Funds

Hedge funds can be seen as a more 'extreme' version of AM. Instead of only taking Long-only views on different asset classes, hedge funds tend to trade on shorter time-frames and take both long and short views on the security of interest.

What do I mean by short? In general, the sentiment of going short on a stock means you believe that you believe TSLA's share price is going to drop. In order to profit from this drop, I go to my broker and borrow shares of TSLA and sell them on the market. Assuming all goes well and the stock price drops, I can then buy back the TSLA's share price at a lower price and then return the stock back to my broker – in effect profiting on the difference in share price. This process is called shorting a stock.

While this might seem like a good strategy in practice, it is inherently riskier than simply going long on a company because if things don't pan out as you hope and the stock price rises, there is no limit as to how high the stock can rise. A good example is the meme stock rally that happened with Gamestop. Basically, a group of reddit traders realised that many hedge funds had decided to short Gamestop stock (Ticker: GME) as explained above. They then decided to team up to buy large quantities of GME which caused the share price to spike. Now, because these hedge funds had borrowed the shares at a specific price without actually owning the stock, they ran the risk of getting "short squeezed" if the share price went above the price they chose to borrow it at. As a result of the spike, many hedge funds had to buy the stock to return it to their broker at a loss - which again caused the share price to continue rising by adding upwards pressure on the stock. As you might have figured out by now, it can be seen that shorting a stock is much riskier because there is no ceiling/limit to how high the share price can rise, so theoretically your loss is unlimited. This is contrary to if you simply owned a stock where the maximum amount you could lose is the money you invested originally, assuming the stock price goes to £0.

There are many different kinds of hedge funds and each Portfolio manager, the chief manager of a pool of money, at the same hedge fund operates differently. There are long/short equity hedge funds, hedge funds that only invest in a particular industry or with a certain investment theme, and hedge funds that take a discretionary/systematic approach. A discretionary approach lets the portfolio manager decide whether or not to buy or hold a given security while a systematic approach utilises a set of predefined rules and price limits set by the portfolio manager that determine whether or not to buy or sell a security. To highlight the difference between both approaches let's look at an example:

Let's say Apple (stock ticker: APPL) has reported huge sales growth over the past year of its iPhones and services. Jane is a systematic portfolio manager while Adam is a discretionary portfolio manager. Both of them decide to buy APPL at a price of \$100 and believe that at that price, APPL is a good investment. Jane sets a lower price limit to sell APPL if it drops below \$85 to cut losses and an upper price limit of \$120 to sell and take profit, while Adam believes from his analysis, APPL will rise to \$120. A week later, the Central Bank announces that it will be raising interest rates so, as explained earlier, the cost of borrowing for corporates is increasing. This results in a widespread correction in the market (i.e. stock prices drop) and the price of APPL plummets to \$80. At this point, Jane has decided to sell APPL because the stock has crossed Jane's cut loss price of \$85. Adam, on the other hand, still believes that AAPL is still the same company fundamentally and will still rise to \$120 - its management is still top notch, business opportunities are still the same and the company has enough cash on hand to weather the higher interest rate environment. He therefore continues holding the stock in the hope it will reach \$120 even though the price has dropped to \$85.

Hedge funds can also either be family offices, where a small group of traders typically trade the founder's money or they could be non-family offices, trading on investor's money – in this case, because of their aggressive strategies, hedge funds typically ask for a 2% base fee along with 20% commission on profits (2 and 20 rule), so for example, if an investor has £8,000 originally invested that has grown to £10,000 with the hedge fund, the hedge fund will take a base fee of £200 (2%) and a further 20% commission on any profits made, leaving 80% to the investor.

Private Equity Funds

PE funds, also referred to as sponsors, are companies that look to acquire another company through a mixture of debt and equity (i.e. through taking loans, issuing bonds and using investor's money) and improve the business to generate earnings to pay off the debt. Typically, 50-70% of the acquisition is paid for through debt.

Like all funds, these companies pull money from investors – these investors could be pension funds, wealthy individuals etc, and invest these funds in companies (equity). The rest of the capital needed to acquire a company is raised through debt – a mixture of loans from banks and the release of bonds to public markets.

The key metric PE funds look at is their Internal Rate of Return (IRR), the rate at which the firm's invested capital is expected to grow each year that the fund owns the company. Typically the return has to be at least 20-30% over five years. After this period, the PE fund can exit their position by either:

1. Selling the company to another PE fund.
2. Selling the company to a strategic buyer (which would be another corporate).
3. IPO-ing the company onto a public exchange.

Examples of prominent PE funds include Blackstone, KKR, CVC, Carlyle Group etc.

Bibliography

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